

Decision Tree Algorithm Assignment

K.N.Toosi University of Technology Introduction
to Data Mining

Fall 2024

Implementation Assignment

Dataset

The dataset for this assignment is the COVID-19 dataset available on Kaggle: [COVID-19 Dataset](#).

Task

Implement a decision tree classification algorithm in Python and compare it to the Sklearn implementation. Unlike Sklearn, do not use one-hot encoding; manually split multi-label categorical features. Compare your performance with the Sklearn decision tree. Performance should be measured via F1-score.

F1-score

The F1-score is the harmonic mean of precision and recall. It is calculated as

follows: where:
$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

$$Precision = \frac{TruePositives}{TruePositives + FalsePositives}$$

$$Recall = \frac{TruePositives}{TruePositives + FalseNegatives}$$

Note

Any attempt to use AI tools for generating the code is strictly prohibited. Students will be asked to present and explain their code during a class session.